ahead of time. A PSK can be used to avoid the overheads of public key cryptography on constrained devices.

In TLS 1.2 and DTLS 1.2, a PSK can be used by specifying dedicated PSK cipher suites such as TLS_PSK_WITH_AES_128_CCM. In TLS 1.3, and the upcoming DTLS 1.3, use of a PSK is instead negotiated using an extension that the client sends in the initial ClientHello message. Once a PSK cipher suite has been selected, the server and client derive session keys from the PSK and random values that they each contribute during the handshake, ensuring that unique keys are still used for every session. The session key is used to compute a HMAC tag over all of the handshake messages, providing authentication of the session: only somebody with access to the PSK could derive the same HMAC key and compute the correct authentication tag.

> **CAUTION** Although unique session keys are generated for each session, the basic PSK cipher suites lack forward secrecy: an attacker that compromises the PSK can easily derive the session keys for every previous session if they captured the handshake messages. Section 12.2.4 discusses PSK cipher suites with forward secrecy.

Because PSK is based on symmetric cryptography, with the client and server both using the same key, it provides mutual authentication of both parties. Unlike client certificate authentication, however, there is no name associated with the client apart from an opaque identifier